

UNIT 1 INTRODUCTION

- “**Mining**” is the process of extraction of some valuable material from the earth e.g. coal mining, diamond mining etc
- **Gregory Piatetsky-Shapiro** coined the term “Knowledge Discovery in Databases” in 1989.
- Currently, Data Mining and Knowledge Discovery are used interchangeably.

Data Mining:

- The process of **extracting useful information from a huge amount of data.**
- Data mining is used by humans to discover new, accurate, and useful patterns in data or meaningful relevant information for the ones who need it.
- Data Mining is also called **Knowledge Discovery of Data (KDD).**
- Data Mining is a process used by organizations to **extract specific data from huge databases** to solve business problems. It primarily turns raw data into useful information.
- Data mining (Knowledge discovery from data)
- Alternative names

Knowledge discovery in databases (KDD), Knowledge extraction, data/pattern analysis, information harvesting, etc.

It is a multi-disciplinary skill that uses machine learning, statistics, AI and database technology

- **WHY DATA MINING**

Volume of information is increasing every day.

Performance of data mining

Data mining can be performed on the following types of data:

Relational Database:

A relational database is a collection of multiple data sets formally organized by tables, records, and columns from which data can be accessed in various ways without having to recognize the database tables.

Data warehouses:

- A Data Warehouse is the technology that collects the data from various sources within the organization to provide meaningful business insights.
- The data warehouse is designed for the analysis of data rather than transaction processing.

Data Repositories:

- The Data Repository generally refers to a **destination for data storage**. For example, a group of databases, where an organization has kept various kinds of information.

Object-Relational Database:

A combination of an object-oriented database model and relational database model is called an object-relational model. It supports Classes, Objects, Inheritance, etc.

Transactional Database:

A transactional database refers to a database management system (DBMS) that has the potential to undo a database transaction if it is not performed appropriately.

- **Knowledge Discovery in Databases (KDD)**-is a multistep process of finding useful information and patterns in data while Data Mining is one of the steps in KDD of using algorithms for extraction of patterns.

- **Steps in KDD**

- Selection
- Preprocessing
- Data Mining
- Pattern Interpretation/evaluation
- Visualization

- **Selection**

- Data Extraction -Obtaining Data from heterogeneous data sources - Databases, Data warehouses, World wide web or other information repositories.

- **Preprocessing**

- Data Cleaning- Incomplete, noisy, inconsistent data to be cleaned Missing data may be ignored or predicted, erroneous data may be deleted or corrected

- **Transformation**

- Data Integration- Combines data from multiple sources into a coherent store -Data can be encoded in common formats, normalized, reduced.

- **Data mining**

- Apply algorithms to transformed data an extract pattern

- **Pattern Interpretation/evaluation**

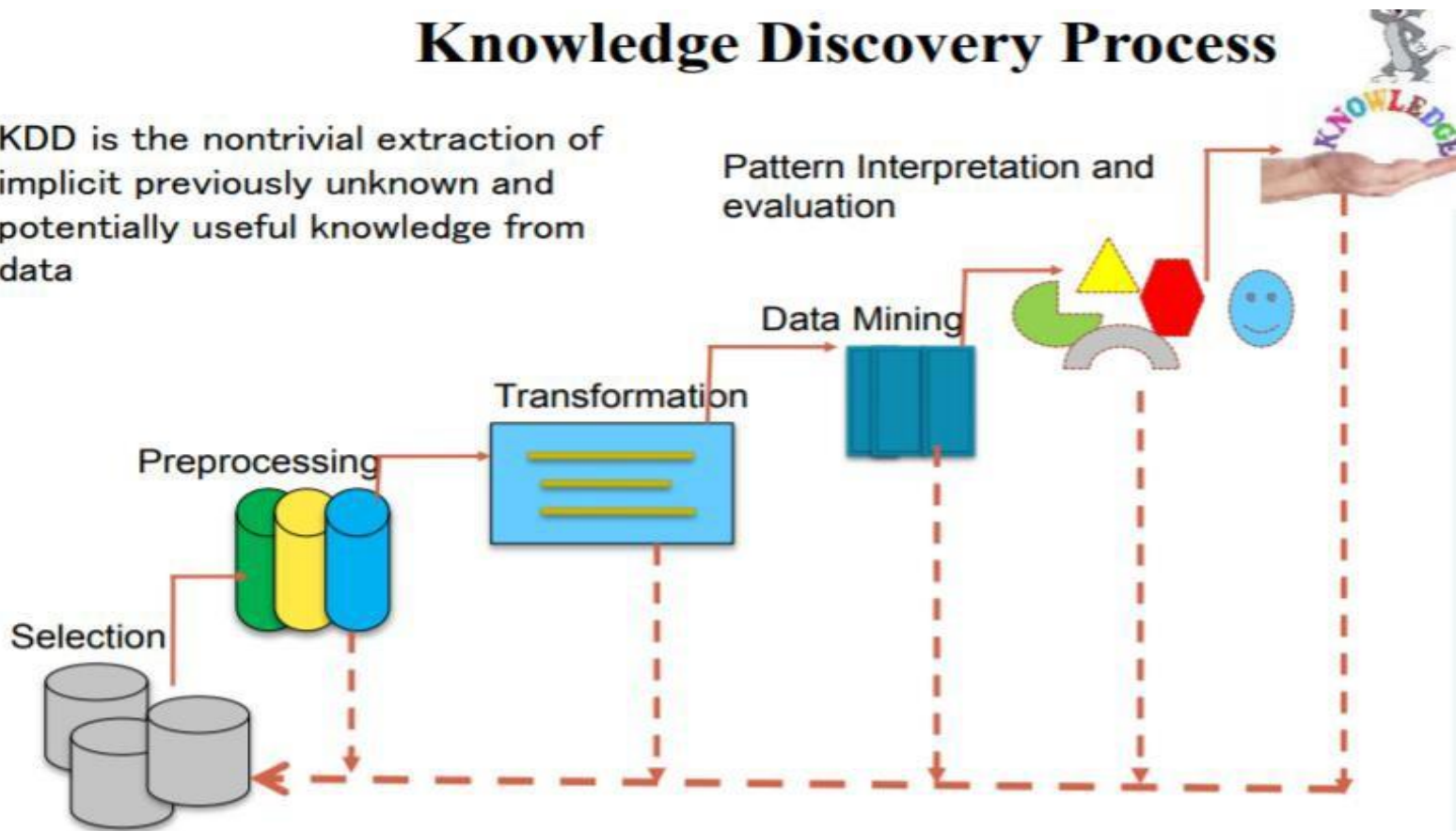
- Pattern Evaluation- Evaluate the interestingness of resulting patterns or apply interestingness measures to filter out discovered patterns

- **Knowledge presentation**

- Present the mined knowledge visualization techniques can be us.

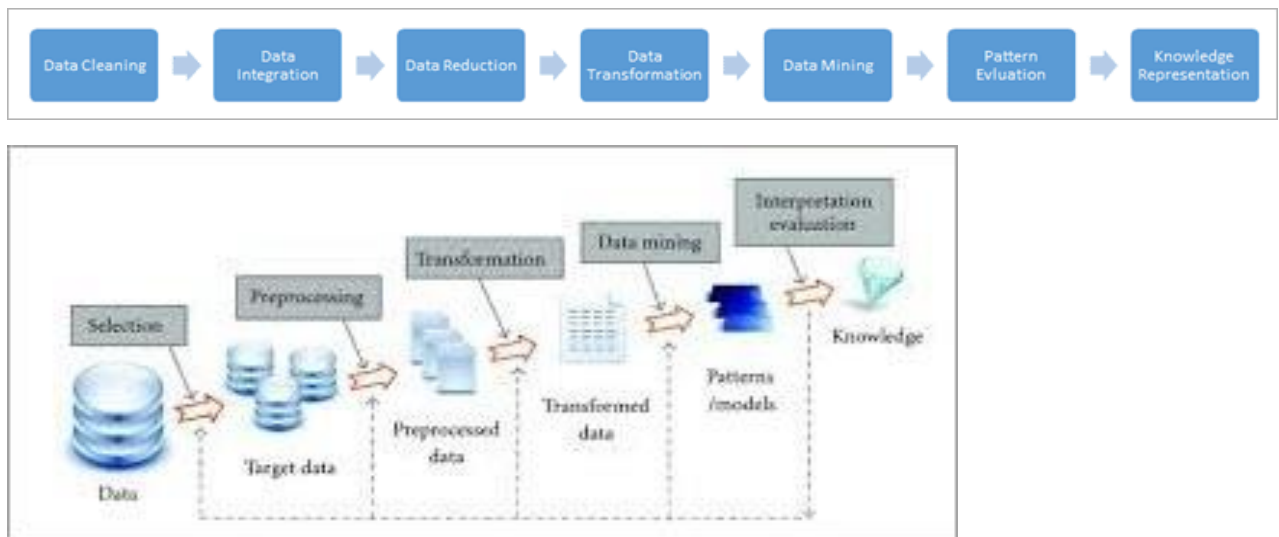
Knowledge Discovery Process

KDD is the nontrivial extraction of implicit previously unknown and potentially useful knowledge from data



Steps In The Data Mining Process

The **data mining process** is divided into two parts i.e. Data Preprocessing and Data Mining. **Data Preprocessing** involves data cleaning, data integration, data reduction, and data transformation. The **data mining** part performs data mining, pattern evaluation and knowledge representation of data.



Why do we preprocess the data?

There are many factors that determine the usefulness of data such as accuracy, completeness, consistency, timeliness. The data has to quality if it satisfies the intended purpose. Thus, preprocessing is crucial in the data mining process. The major steps involved in data preprocessing are explained below.

1) Data Cleaning

Data cleaning is the first step in data mining. It holds importance as dirty data if used directly in mining can cause confusion in procedures and produce inaccurate results.

Basically, this step involves the removal of noisy or incomplete data from the collection. Many methods that generally clean data by itself are available but they are not robust.

This step carries out the routine cleaning work by:

(i) Fill The Missing Data:

Missing data can be filled by methods such as:

- Ignoring the tuple.
- Filling the missing value manually.
- Use the measure of central tendency, median or
- Filling in the most probable value.

(ii) Remove The Noisy Data: Random error is called noisy data.

Methods to remove noise are:

Binning: Binning methods are applied by sorting values into buckets or bins. Smoothing is performed by consulting the neighboring values.

Binning is done by smoothing by bin i.e. each bin is replaced by the mean of the bin. Smoothing by a median, where each bin value is replaced by a bin median. Smoothing by bin boundaries i.e. The minimum and maximum values in the bin are bin boundaries and each bin value is replaced by the **closest boundary value**.

2) Data Integration

When multiple heterogeneous data sources such as databases, data cubes or files are combined for analysis, this process is called data integration. This can help in improving the accuracy and speed of the data mining process.

Different databases have different naming conventions of variables, by causing redundancies in the databases. Additional Data Cleaning can be performed to remove the redundancies and inconsistencies from the data integration without affecting the reliability of data.

Data Integration can be performed using Data Migration Tools such as Oracle Data Service Integrator and Microsoft SQL etc.

3) Data Reduction

This technique is applied to obtain relevant data for analysis from the collection of data. The size of the representation is much smaller in volume while maintaining integrity. **Data Reduction is performed using methods such as Naive Bayes, [Decision Trees](#), Neural network, etc.**

Some strategies of data reduction are:

- **Dimensionality Reduction:** Reducing the number of attributes in the dataset.
- **Numerosity Reduction:** Replacing the original data volume by smaller forms of data representation.
- **Data Compression:** Compressed representation of the original data.

4) Data Transformation

In this process, data is transformed into a form suitable for the data mining process. Data is consolidated so that the mining process is more efficient and the patterns are easier to understand. Data Transformation involves Data Mapping and code generation process.

Strategies for data transformation are:

- **Smoothing:** Removing noise from data using clustering, regression techniques, etc.
- **Aggregation:** Summary operations are applied to data.
- **Normalization:** Scaling of data to fall within a smaller range.
- **Discretization:** Raw values of numeric data are replaced by intervals. **For Example,** Age.

5) Data Mining

Data Mining is a process to identify interesting patterns and knowledge from a large amount of data. In these steps, intelligent patterns are applied to extract the data patterns. The data is represented in the form of patterns and models are structured using classification and clustering techniques.

6) Pattern Evaluation

This step involves identifying interesting patterns representing the knowledge based on interestingness measures. **Data summarization and visualization methods** are used to make the data understandable by the user.

7) Knowledge Representation

Knowledge representation is a step where data visualization and knowledge representation tools are used to represent the mined data. Data is visualized in the form of reports, tables, etc.

Tasks and Functionalities of Data Mining:

Data mining tasks can be classified into two types: descriptive and predictive. Descriptive mining tasks define the common features of the data in the database, and the predictive mining tasks act in inference on the current information to develop predictions.

Data mining activities can be divided into two categories:

- **Descriptive Data Mining:** It includes certain knowledge to understand what is happening within the data without a previous idea. The common data features are highlighted in the data set. For example, count, average etc.
- **Predictive Data Mining:** It helps developers to provide unlabeled definitions of attributes. With previously available or historical data, data mining can be used to make predictions about critical business metrics based on data's linearity. For example, predicting the volume of business next quarter based on performance in the previous quarters over several years or judging from the findings of a patient's medical examinations that is he suffering from any particular disease.

Data Mining Functionalities is to observe the various trends in data mining. There are several data mining functionalities that the organized and scientific methods offer, such as:

- **Data characterization** – It is a summarization of the general characteristics of an object class of data. The data corresponding to the user-specified class is generally collected by a database query. The output of data characterization can be presented in multiple forms.

- **Data discrimination** – It is a comparison of the general characteristics of target class data objects with the general characteristics of objects from one or a set of contrasting classes. The target and contrasting classes can be represented by the user, and the equivalent data objects fetched through database queries.
- **Association Analysis** – It analyses the set of items that generally occur together in a transactional dataset. There are two parameters that are used for determining the association rules –
 - It provides which identifies the common item set in the database.
 - Confidence is the conditional probability that an item occurs in a transaction when another item occurs.
- **Classification** – Classification is the procedure of discovering a model that represents and distinguishes data classes or concepts, for the objective of being able to use the model to predict the class of objects whose class label is anonymous. The derived model is established on the analysis of a set of training data (i.e., data objects whose class label is common).
- **Prediction** – It defines predict some unavailable data values or pending trends. An object can be anticipated based on the attribute values of the object and attribute values of the classes. It can be a prediction of missing numerical values or increase/decrease trends in time-related information.
- **Clustering** – It is similar to classification but the classes are not predefined. The classes are represented by data attributes. It is unsupervised learning. The objects are clustered or grouped,

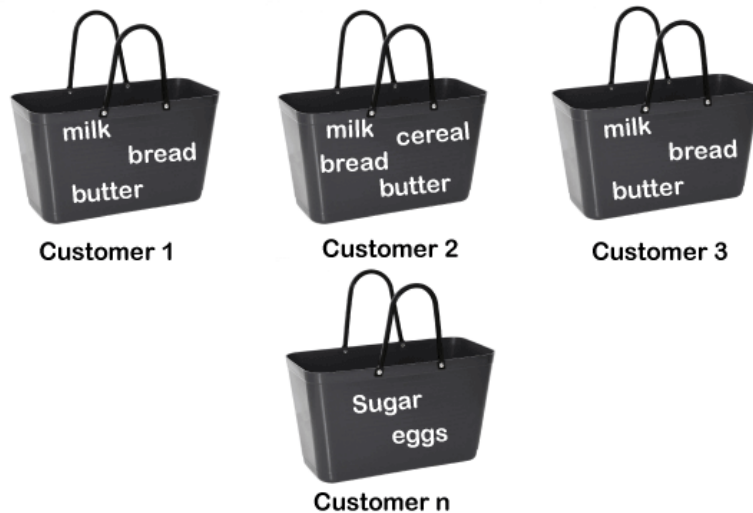
depends on the principle of maximizing the intraclass similarity and minimizing the intraclass similarity.

- **Outlier analysis** – Outliers are data elements that cannot be grouped in a given class or cluster. These are the data objects which have multiple behaviour from the general behaviour of other data objects. The analysis of this type of data can be essential to mine the knowledge.
- **Evolution analysis** – It defines the trends for objects whose behaviour changes over some time.

Association Rule Learning

- Association rule learning is a type of **unsupervised learning** technique that **checks for the dependency of one data item on another data item** and maps accordingly so that it can be more profitable.
- It tries to **find some interesting relations or associations** among the variables of dataset.
- It is based on different **rules to discover the interesting relations** between variables in the database.
- The association rule learning is one of the very important concepts of [machine learning](#), and it is employed in **Market Basket analysis, Web usage mining, continuous production, etc.** Here market basket analysis is a technique used by the various big retailer to discover the associations between items.

For example, if a customer buys bread, he most likely can also buy butter, eggs, or milk, so these products are stored within a shelf or mostly nearby. Consider the below diagram:



Association rule learning can be divided into three types of algorithms:

1. **Apriori**
2. **Eclat**
3. **F-P Growth Algorithm**

How does Association Rule Learning work?

Association rule learning works on the **concept of If and Else Statement**, such as if A then B.



Association Rule Learning

Here the **If element is called antecedent**, and **then statement is called as Consequent**. These types of relationships where we can find out some association or relation between two items is known as **single cardinality**. It is all about creating rules, and if the number of

items increases, then cardinality also increases accordingly. So, to measure the associations between thousands of data items, there are several metrics.

- **Support**
- **Confidence**
- **Lift**

Support

Support is the frequency of A or how frequently an item appears in the dataset. It is defined as the fraction of the transaction T that contains the itemset X. If there are X datasets, then for transactions T, it can be written as:

$$\text{Supp}(X) = \frac{\text{Freq}(X)}{T}$$

Confidence

Confidence indicates how often the rule has been found to be true. Or how often the items X and Y occur together in the dataset when the occurrence of X is already given. It is the ratio of the transaction that contains X and Y to the number of records that contain X.

$$\text{Confidence} = \frac{\text{Freq}(X,Y)}{\text{Freq}(X)}$$

Lift

It is the strength of any rule, which can be defined as below formula:

$$\text{Lift} = \frac{\text{Supp}(X,Y)}{\text{Supp}(X) \times \text{Supp}(Y)}$$

It is the ratio of the observed support measure and expected support if X and Y are independent of each other. It has three possible values:

- If **Lift**= 1: The probability of occurrence of antecedent and consequent is independent of each other.
- **Lift**>1: It determines the degree to which the two itemsets are dependent to each other.
- **Lift**<1: It tells us that one item is a substitute for other items, which means one item has a negative effect on another.

Types of Association Rule Learning

Association rule learning can be divided into three algorithms:

Apriori Algorithm

This algorithm uses frequent datasets to generate association rules. It is designed to work on the databases that contain transactions. This algorithm uses a breadth-first search and Hash Tree to calculate the itemset efficiently.

It is mainly used for market basket analysis and helps to understand the products that can be bought together. It can also be used in the healthcare field to find drug reactions for patients.

Eclat Algorithm

Eclat algorithm stands for Equivalence Class Transformation. This algorithm uses a depth-first search technique to find frequent itemsets in a transaction database. It performs faster execution than Apriori Algorithm.

F-P Growth Algorithm

The F-P growth algorithm stands for Frequent Pattern, and it is the improved version of the Apriori Algorithm. It represents the database in the form of a tree structure that is known as a frequent pattern or tree. The purpose of this frequent tree is to extract the most frequent patterns.

Applications of Association Rule Learning

It has various applications in machine learning and data mining. Below are some popular applications of association rule learning:

- **Market Basket Analysis:** It is one of the popular examples and applications of association rule mining. This technique is commonly used by big retailers to determine the association between items.
- **Medical Diagnosis:** With the help of association rules, patients can be cured easily, as it helps in identifying the probability of illness for a particular disease.
- **Protein Sequence:** The association rules help in determining the synthesis of artificial Proteins.
- It is also used for the **Catalog Design** and **Loss-leader Analysis** and many more other applications.